

Python for DevOps - 7 Day Intensive Bootcamp

Goal

7 din ke end tak tum Python me automation scripts likh sako jo Git, Docker, Kubernetes aur Terraform ke commands execute kar sake.

Daily Routine (4–5 Hours)

Session 1 (45 Minutes)

Theory

Session 2 (90 Minutes)

Code Along

Session 3 (90 Minutes)

Practice Questions

Session 4 (60–90 Minutes)

Mini Automation Project

Rule:

- Har code khud type karo.
 - Copy-paste mat karo.
 - Har din GitHub par code push karo.
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Day 1 - Python Foundation

Topics

- Python Installation
- VS Code Setup
- Variables
- Data Types
- Input
- Output
- Operators
- If Else

- Basic Debugging

Practice (15 Programs)

1. Hello World
2. Print Employee Details
3. Add Two Numbers
4. Calculator
5. Even/Odd
6. Largest Number
7. Age Checker
8. Celsius to Fahrenheit
9. Kilometer to Meter
10. Salary Calculator
11. Voting Eligibility
12. Discount Calculator
13. Percentage Calculator
14. Swap Two Variables
15. BMI Calculator

Mini Project: Interactive Calculator

Day 2 - Loops and Functions

Topics

- for
- while
- break
- continue
- range
- Functions
- return
- Default Arguments

Practice (15 Programs)

1. Multiplication Table
2. Factorial
3. Prime Number
4. Fibonacci
5. Armstrong Number
6. Palindrome
7. Reverse Number
8. Reverse String
9. Sum of Digits
10. Count Digits
11. Star Pattern
12. Pyramid Pattern
13. Number Pattern

- 14. Guess Number Game
- 15. Password Retry Program

Mini Project: Student Result Management

Day 3 - Data Structures

Topics

Strings

Lists

Tuples

Sets

Dictionary

Nested Dictionary

Loop Dictionary

Sorting

Searching

Practice

Employee Database

Student Database

Inventory

Server List

Application Configuration

Mini Project

Server Inventory Manager

Day 4 - Files + JSON + YAML

Topics

Read File

Write File

Append

CSV

JSON

YAML

Path Handling

Practice

Read Log File

Count Errors

Create Report

Read Kubernetes YAML

Read JSON Config

Mini Project

Log Analyzer

Day 5 - Python Modules

Topics

os

pathlib

shutil

datetime

logging

sys

subprocess

argparse

Practice

Create Folder

Rename Folder

Delete Folder

Copy Files

Move Files

Run Terminal Commands

Mini Project

Project Structure Generator

Automatically create

project/

src/

logs/

config/

tests/

docs/

Day 6 - DevOps Automation

Topics

subprocess

Git Commands

Docker Commands

Kubectl Commands

Terraform Commands

Helm Commands

Practice

Run

git status

git pull

docker ps

docker images

kubectl get pods

kubectl get nodes

terraform validate

terraform plan

Mini Project

DevOps Command Runner

User selects menu

Python executes command

Day 7 - Final Project

Build

DEVOPS AUTOMATION TOOLKIT

Menu

1 Git

2 Docker

3 Kubernetes

4 Terraform

5 Logs

6 Exit

Features

Git Status

Git Pull

Docker Images

Docker Containers

Restart Deployment

Terraform Validate

Terraform Plan

Show Pod Logs

Create Folder Structure

Generate Log Report

Save Output to File

Error Handling

Logging

Congratulations

After this project you can start writing automation scripts for your DevOps work.

Libraries to Learn

json

os

pathlib

subprocess

logging

shutil

datetime

requests

PyYAML

csv

Folder Structure

python-devops/

day01/

day02/

day03/

day04/

day05/

day06/

day07/

projects/

notes/

Daily Git Commit Format

Day-01 Python Basics Completed

Day-02 Loops Completed

Day-03 Data Structures Completed

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Resources

Official Python Documentation

Python Tutorial (Official)

Practice on HackerRank or LeetCode (Easy problems only for Week 1)

Final Target

At the end of 7 days you should be able to write scripts that:

- Read and write files
- Parse JSON and YAML
- Execute shell commands
- Automate Git operations
- Automate Docker commands
- Automate Kubernetes commands
- Automate Terraform commands
- Generate logs and reports
- Build simple CLI tools